

UAV Initial Sizing — Design Report

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1 · Mission Parameters

Parameter	Value
Cruise speed	100.0 m/s (360 km/h)
Mission altitude	200 m
Mission range	1000 km
Mission endurance	5.0 hr
Payload role	Strike
Mission objective	Low cost
UAV class	large
Max. load factor	3.50 g
Engine type	Piston

2 · Weight Budget

Component	Mass [kg]
MTOW	1210.7
Empty weight	731.8
Fuel weight	0.0
Payload weight	478.9
Fuel fraction W_f/W_0	0.000
Empty fraction W_e/W_0	0.604

3 · Wing & Aerodynamic Sizing

Parameter	Value
Wing area S	8.96 m ²
Wing span b	8.84 m
Aspect ratio AR	8.71
Wing loading W/S	1324.8 N/m ²
Cruise L/D	9.78
Power loading W/P [kg/W]	0.07712 (13.0 W/kg)

4 · Fuselage

Parameter	Value
Fuselage length	7.12 m
Fuselage radius	0.427 m
Cylinder start	10.0 %

Cylinder end	70.0 %
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5 · Longitudinal Stability

Parameter	Value
CG position (from nose)	2.740 m
Neutral point (from nose)	3.678 m
Static margin SM	8717.4 % MAC
Assessment	STABLE — SM = 87.2% MAC (target 5–15%)

6 · Design Point — W/P vs W/S Diagram

Wing Loading vs Weight-to-Power Across Altitudes

